المعهد الهنـدي للتكنولوجيـــا دلهي
أبـــوظبي
INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

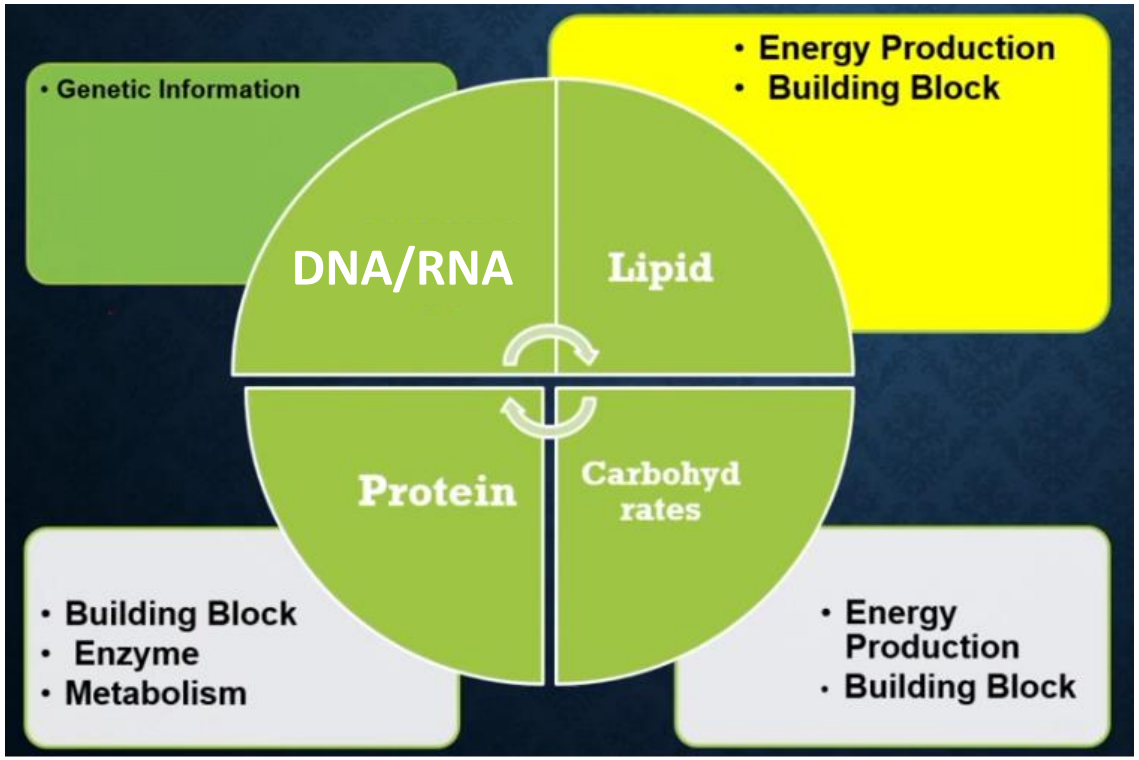
Biomolecules - Carbohydrates

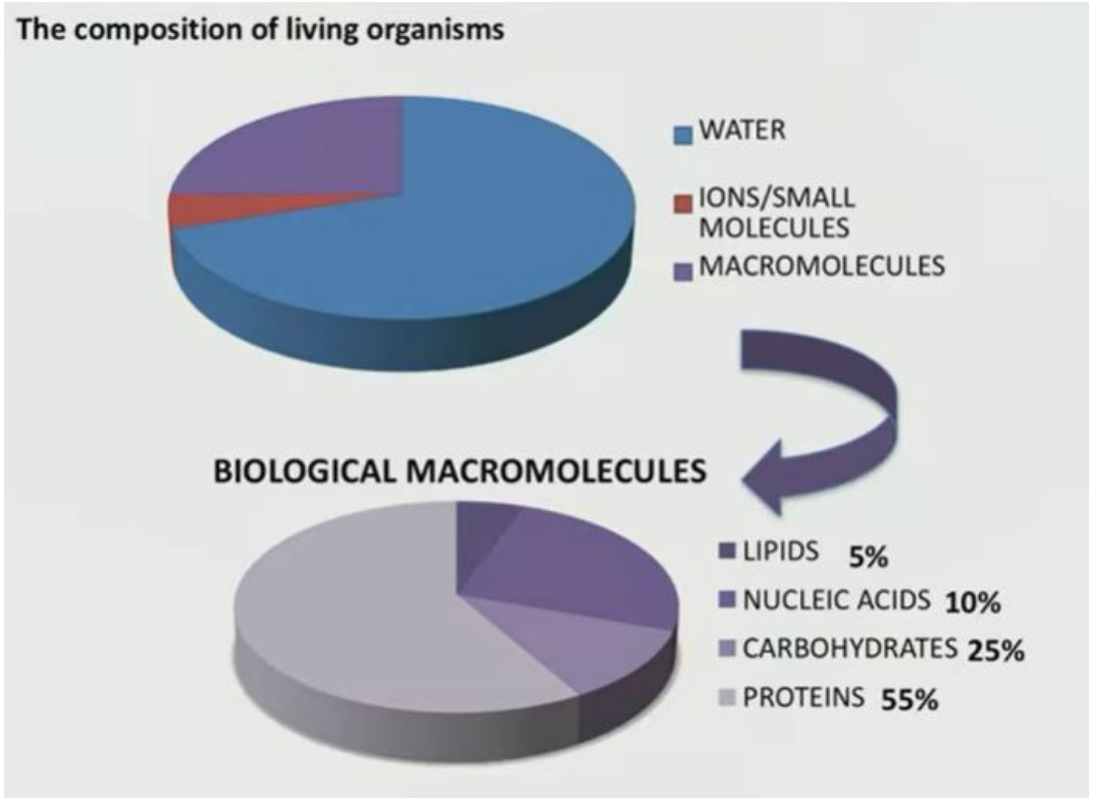
Lecture 9



Prof. Saurabh Raj
Kusuma School of Biological Sciences
IIT Delhi

Contact: *sraj@iitdabudhabi.ac.ae*

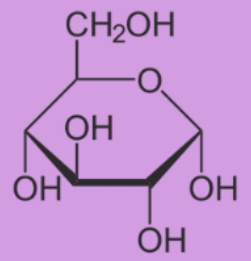

• Genetic Information
• Energy Production
• Building Block
DNA/RNA
Lipid
Protein
Carbohyd rates
• Building Block
• Enzyme
• Metabolism
• Energy Production
• Building Block


The composition of living organisms
WATER
IONS/SMALL MOLECULES
MACROMOLECULES
BIOLOGICAL MACROMOLECULES
LIPIDS 5%
NUCLEIC ACIDS 10%
CARBOHYDRATES 25%
PROTEINS 55%

Examples of Carbohydrates

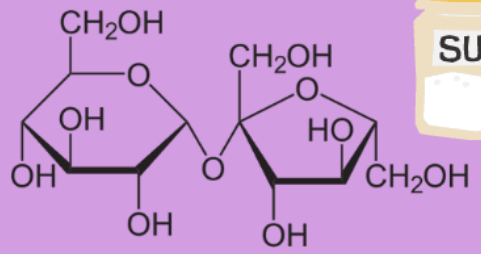
Carbohydrates are organic molecules containing carbon, hydrogen, and oxygen.

Sugars





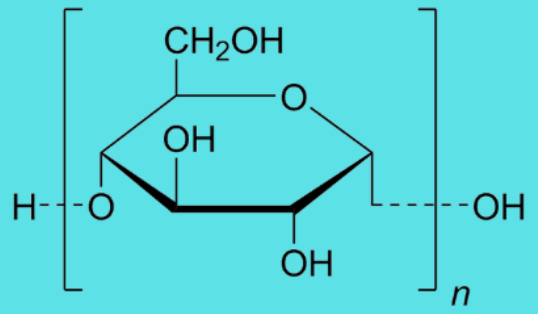
glucose





sucrose (table sugar)

Oligosaccharides



α-1,4

$2 < n < 20$



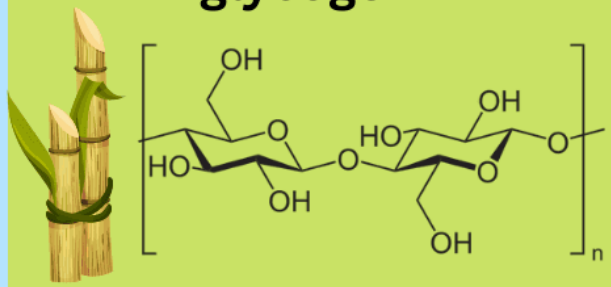


maltodextrin

Polysaccharides



glycogen



cellulose

sciencenotes.org

Carbohydrate



- Carbohydrates include sugars and the polymers of sugars.
- The simplest carbohydrates are monosaccharides, or single sugars.
- Carbohydrate macromolecules are composed of many monosaccharides, polysaccharides.
- It serves as fuel and building material in cells

Carbohydrates: one family, many jobs

Energy

Glucose is oxidized to support ATP production.

Storage

Starch in plants; glycogen in animals.

Structure

Cellulose reinforces plant walls; chitin provides tough support.

Genetic structural support

Forms part of DNA and RNA



Structure determines biological function

How carbohydrates are classified

Monosaccharide



1 unit

glucose, fructose

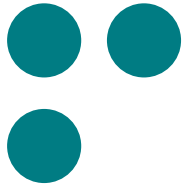
Disaccharide



2 units

sucrose, lactose

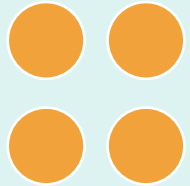
Oligosaccharide



3–10 units

cell-surface tags

Polysaccharide



many units

starch, glycogen

Monosaccharides: carbonyl position + carbon count

Aldose

Carbonyl at the end of the chain
Example: glucose

Ketose

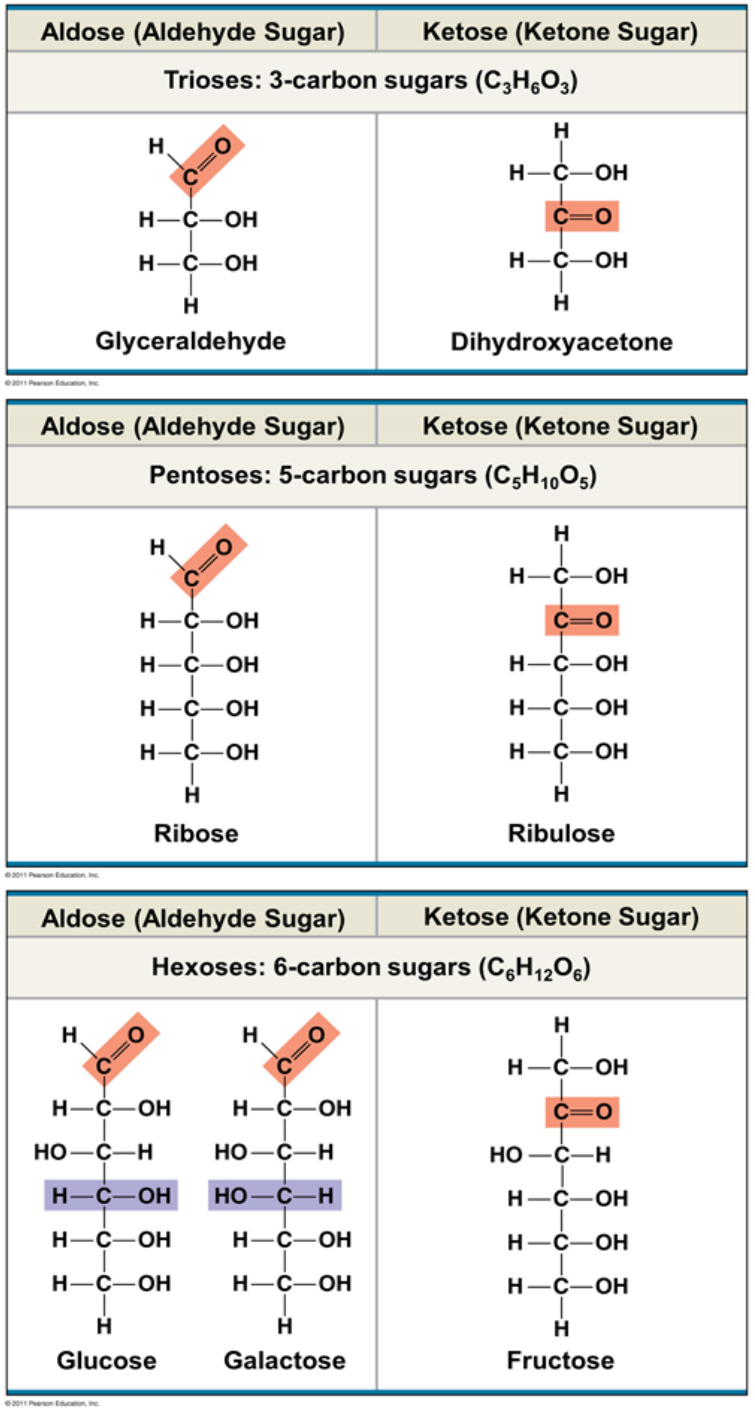
Carbonyl within the chain
Example: fructose

3 carbons	triose	glyceraldehyde
5 carbons	pentose	ribose
6 carbons	hexose	glucose / fructose

Names often encode structure: aldohexose, ketohexose

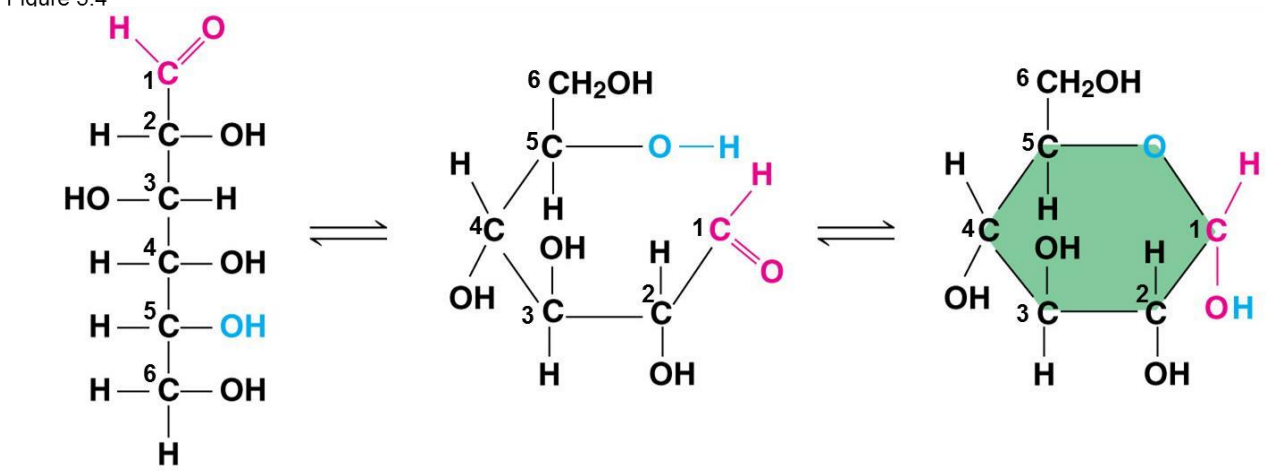
Monosaccharides

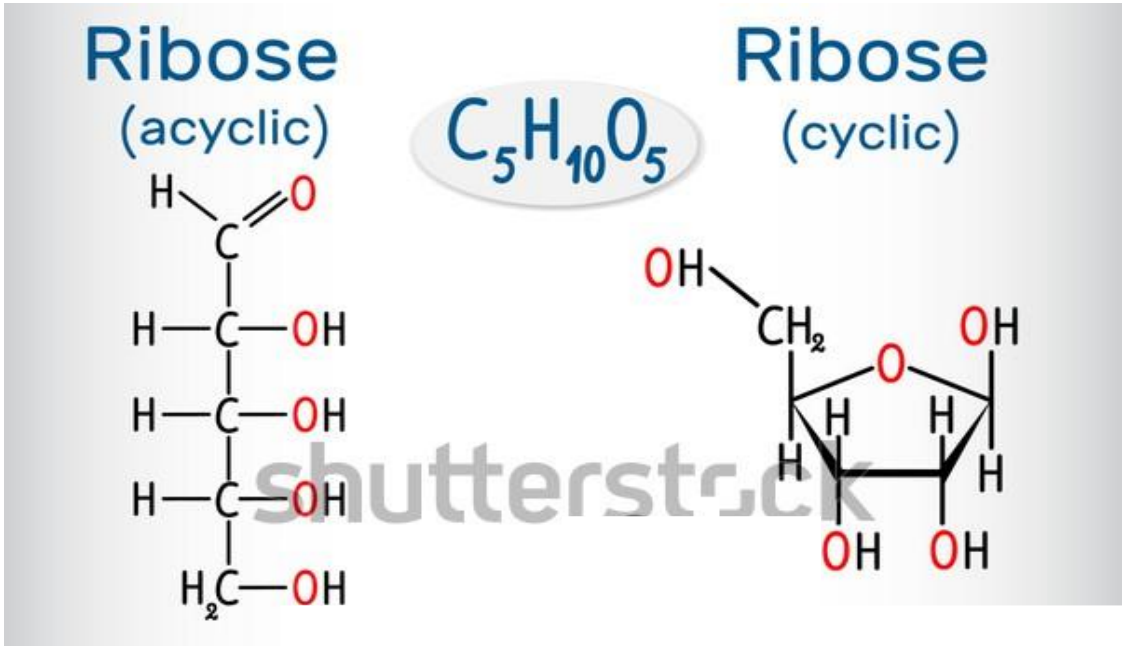
- Monosaccharides have molecular formulas that are usually multiples of CH_2O
- Glucose ($C_6H_{12}O_6$) is the most common monosaccharide.
- Monosaccharides are classified by
  - The location of the carbonyl group (as aldose or ketose)
  - The number of carbons in the carbon skeleton

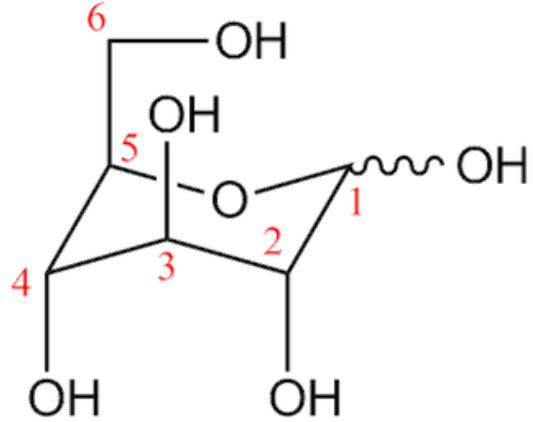


- Though often drawn as linear skeletons, in aqueous solutions many sugars form rings.
- Monosaccharides serve as a major fuel for cells and as raw material for building molecules.

Figure 5.4



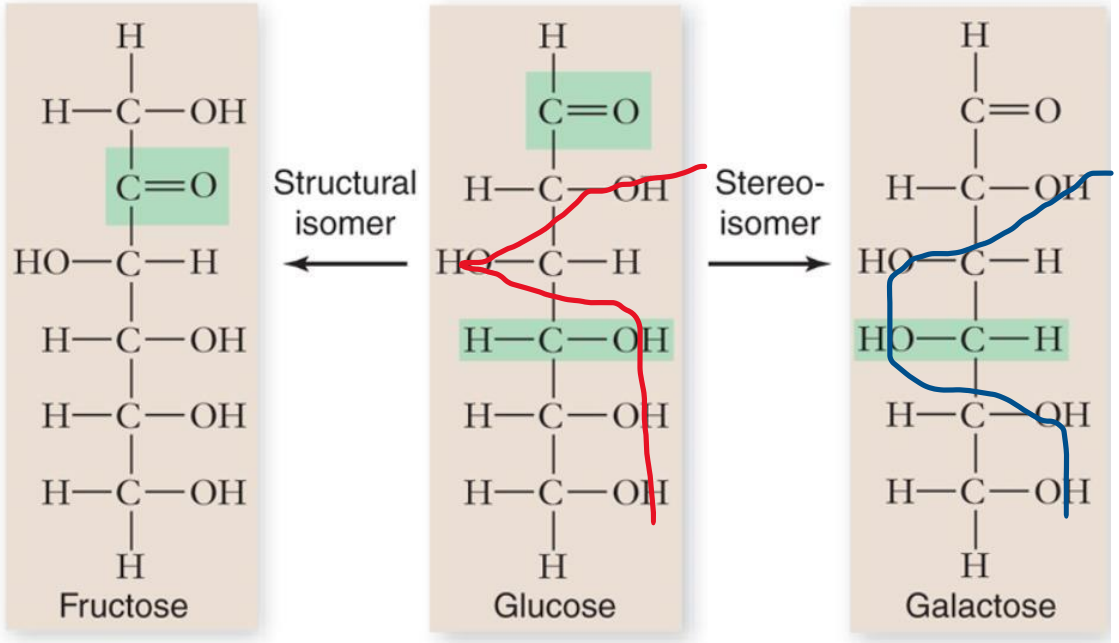




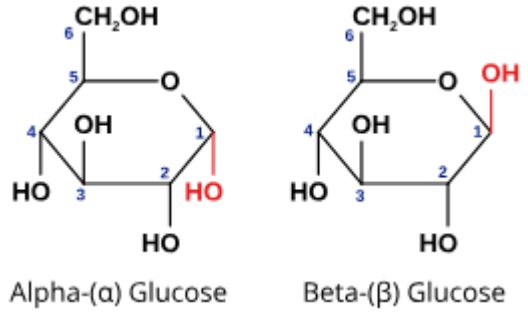
https://www.pearson.com/channels/biology

Isomers: Same molecular formulas but different structures

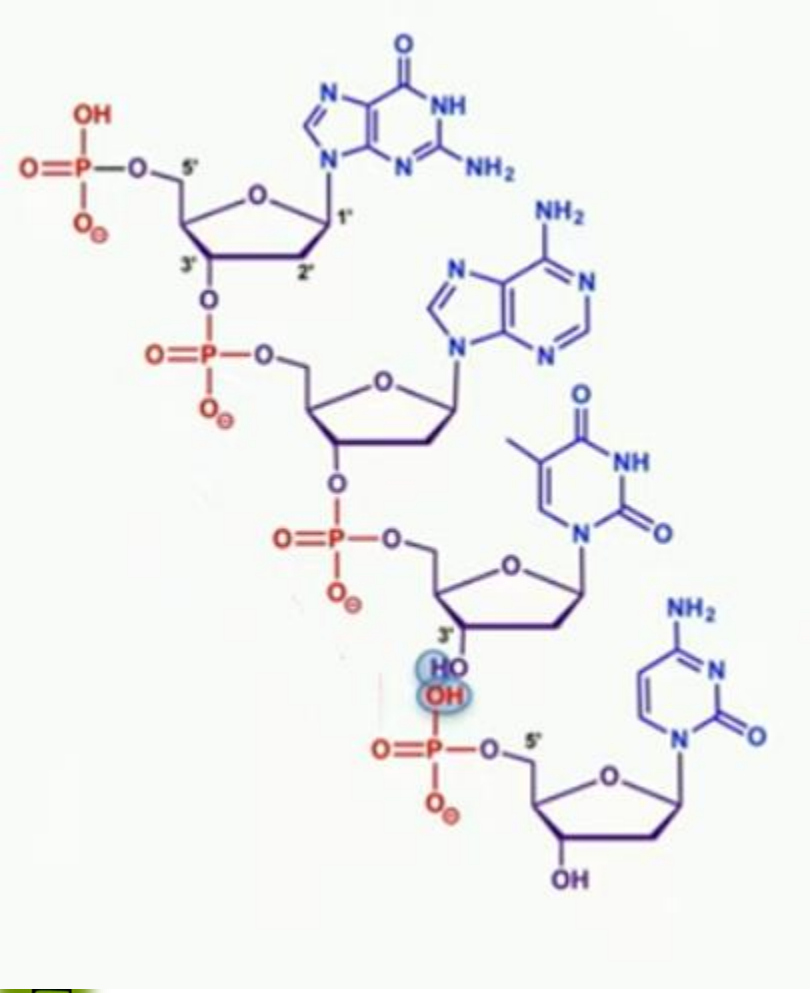
- Structural isomer = difference in the C skeleton structure. Depending on the location of the carboxyl group.
- Stereoisomer = difference in location of functional groups.
  Spatial arrangement of their parts around asymmetric carbons.

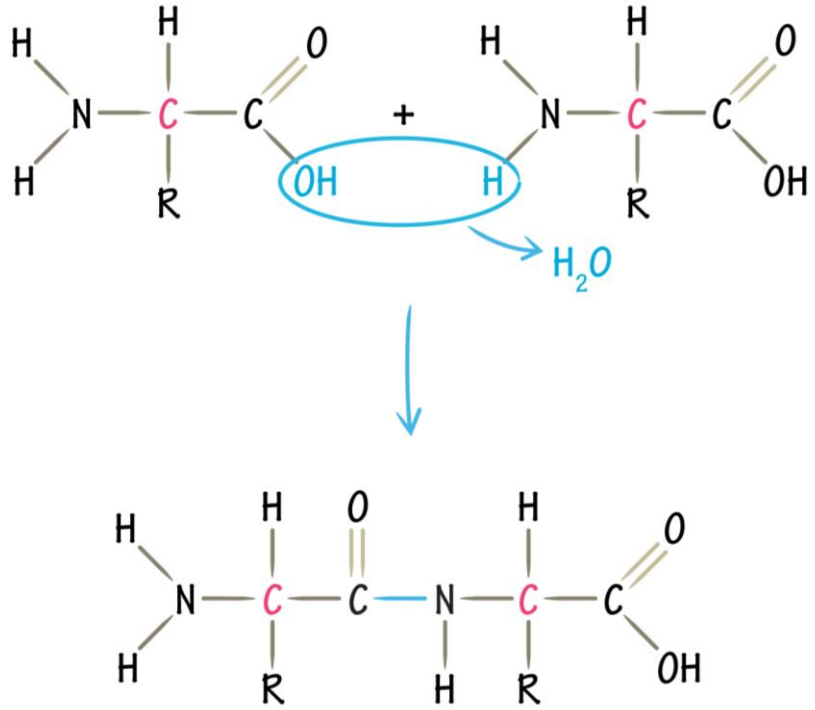


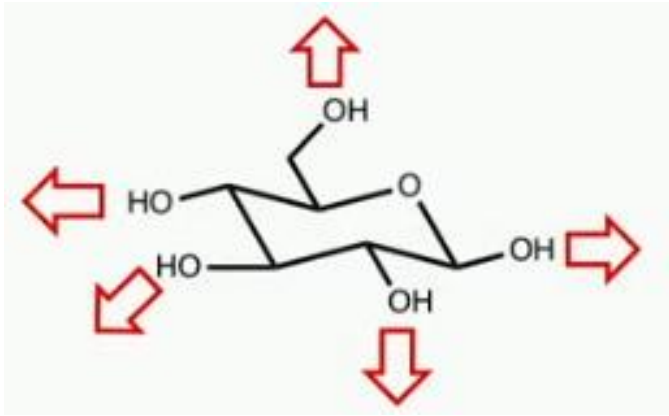
- **Alpha (α) Glucose:** The hydroxyl group on carbon-1 points **downward** below the plane of the ring.
- **Beta (β) Glucose:** The hydroxyl group on carbon-1 points **upward** above the plane of the ring.



Polysaccharide







Disaccharide

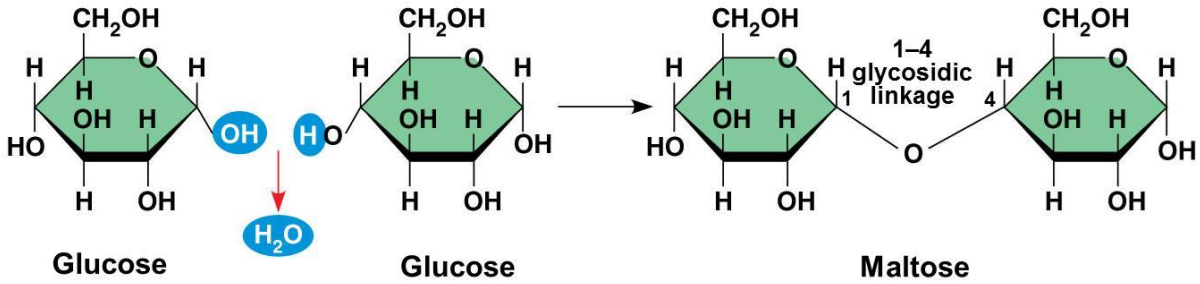
- A ***disaccharide*** is formed when a dehydration reaction joins two monosaccharides. (Condensation)
- This covalent bond is called a glycosidic linkage.

Example:

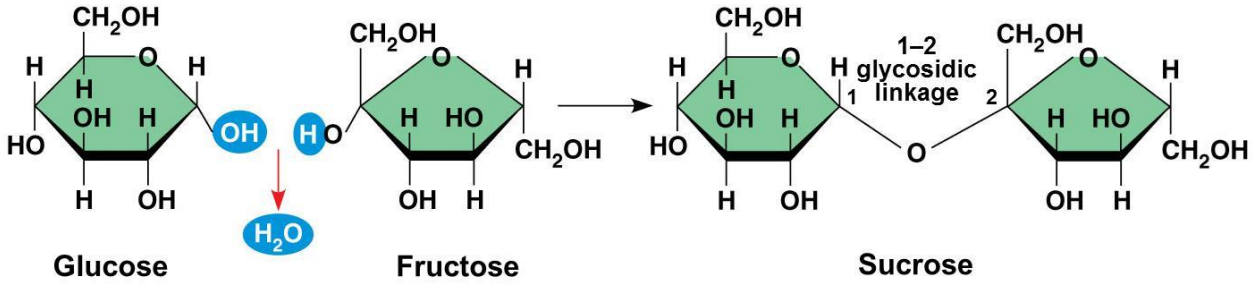
Maltose, linking of two molecules of glucose.

Sucrose, which is table sugar, formed by joining of glucose and fructose.

Lactose, the sugar present in milk, a glucose molecule joined to a galactose molecule.

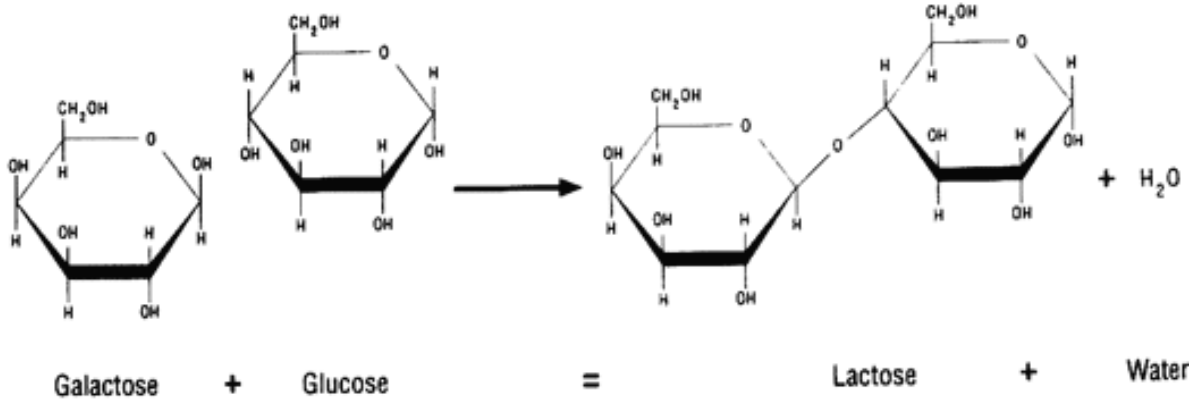


(a) Dehydration reaction in the synthesis of maltose



(b) Dehydration reaction in the synthesis of sucrose

© 2011 Pearson Education, Inc.



Common Disaccharide

Maltose	glucose + glucose	α(1→4)	starch digestion
Lactose	galactose + glucose	β(1→4)	milk sugar
Sucrose	glucose + fructose	α1↔β2	transport sugar in plants

Polysaccharides

- Polysaccharides, the polymers of sugars, have storage and structural roles.
- The structure and function of a polysaccharide are determined by its sugar monomers and the positions of glycosidic linkages.

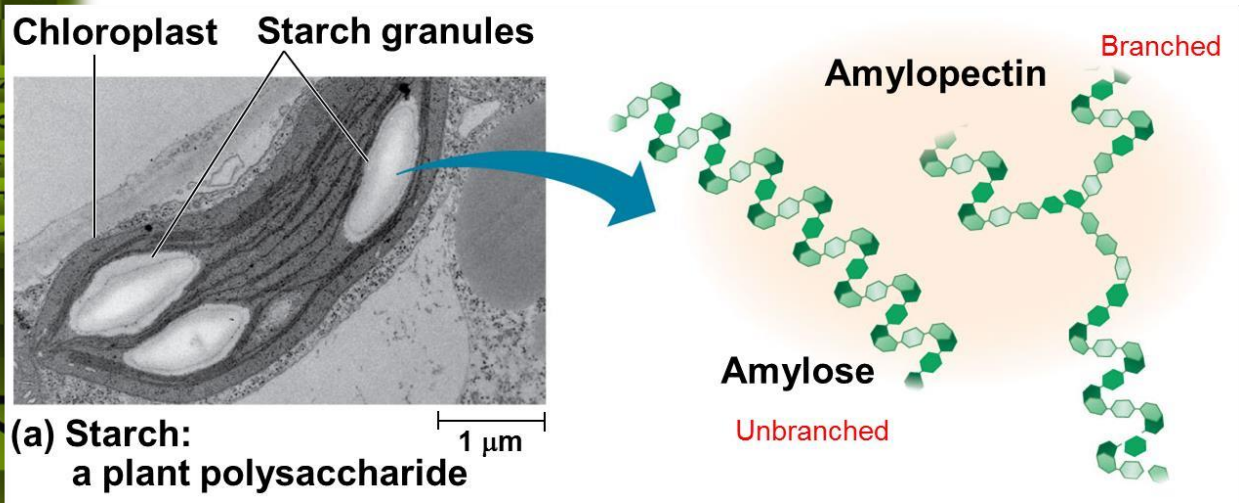
  Starch

  Cellulose

  Glycogen

  Chitin

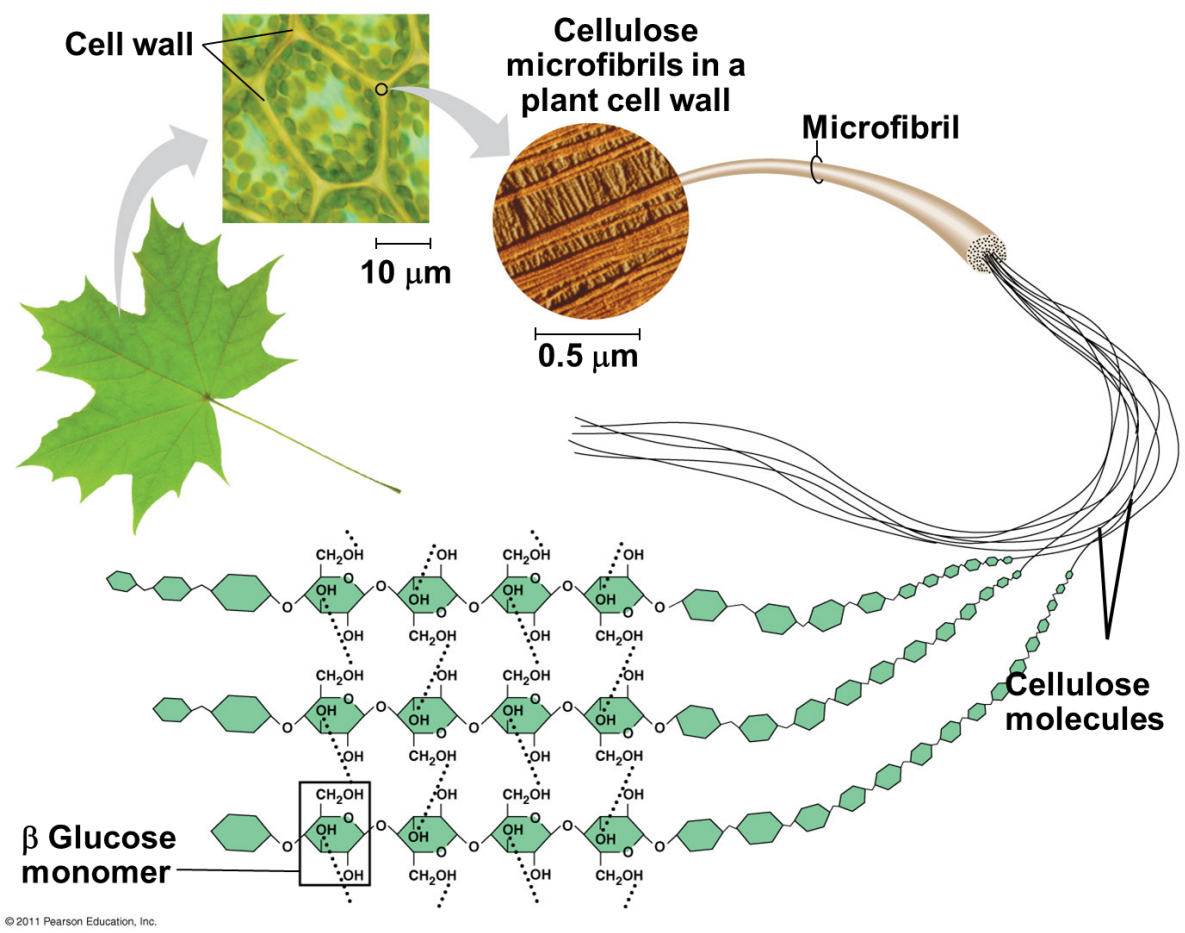
Starch (Storage)



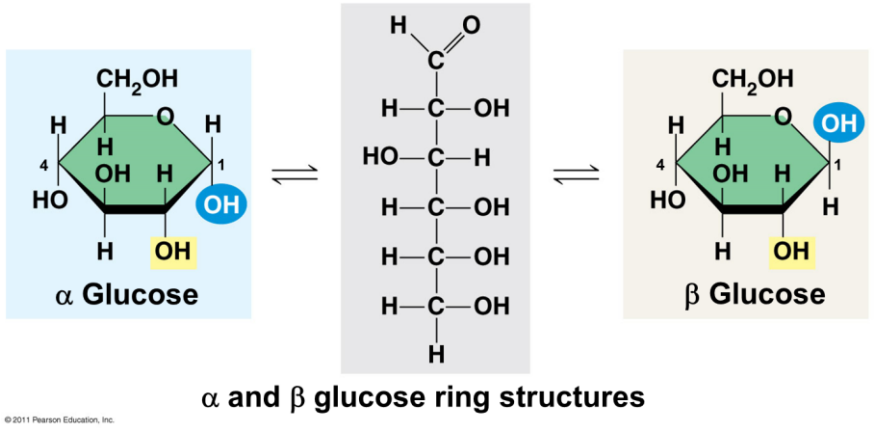
- **Starch**, a storage polymer of glucose monomers.
- It is a complex carbohydrate that exists in many foods, including grains, vegetables, and fruits.
- Stored as granules within cellular structures like plastids, chloroplasts in plants.
- Glucose is a major cellular fuel, starch represents stored energy.
- Bonds between the glucose monomers can be broken by hydrolysis.
- Most animals, including humans, also have enzymes that can hydrolyze plant starch, making glucose available as a nutrient for cells.

Cellulose (Structural)

- Like starch, cellulose is a polymer of glucose, but differ in the glycosidic linkages.
- Because of this small difference, body cannot digest cellulose the same way it can starches & sugars.
- Unlike animal cells, plant cells have a cell wall made of cellulose.
- Human can not digest the cellulose as enzymes that digest it is absent.
- Some microbes, many herbivores use enzymes to digest cellulose.

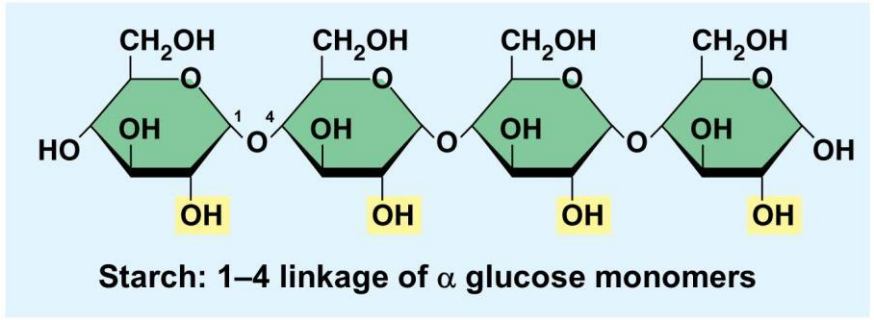


Difference between Starch and Cellulose

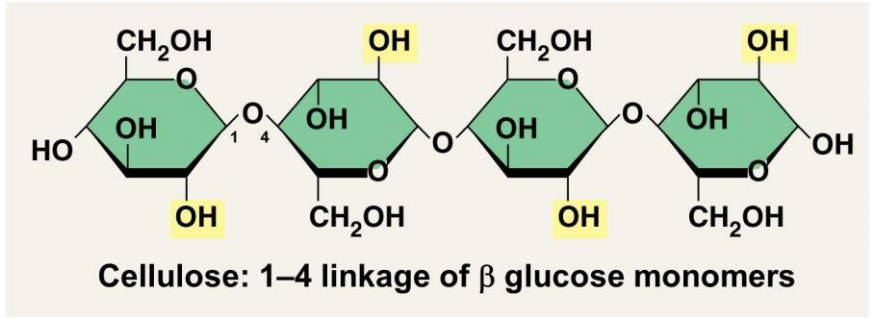


α and β glucose ring structures

© 2011 Pearson Education, Inc.



Starch: 1–4 linkage of α glucose monomers



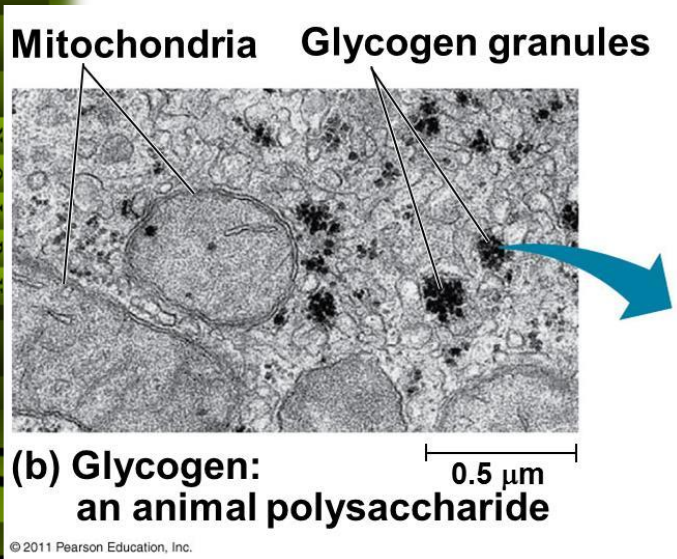
Cellulose: 1–4 linkage of β glucose monomers

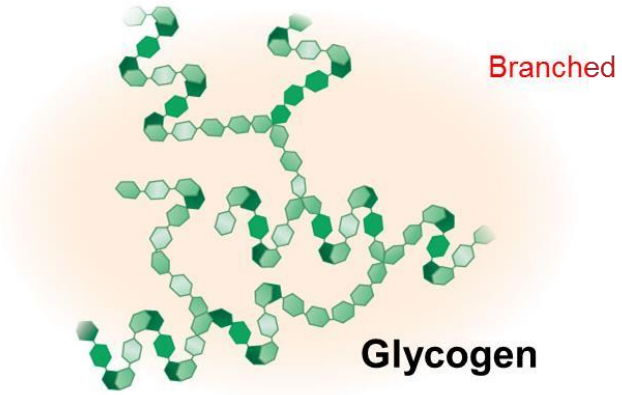
- The difference is based on the fact that glucose has two slightly different ring structures.
- When glucose forms a ring, the hydroxyl group attached to the number 1 carbon is positioned either below or above the plane of the ring.
- These two ring forms for glucose are called alpha (α) and beta (β), respectively.

- Polymers with α glucose are helical.

  Polymers with β glucose are straight.
- In straight structures, H atoms on one strand can bond with OH groups on other strands.
- Parallel cellulose molecules held together this way are grouped into microfibrils, which form strong building materials for plants.
- Enzymes that digest starch by hydrolyzing linkages can't hydrolyze linkages in cellulose.

Glycogen (Storage)

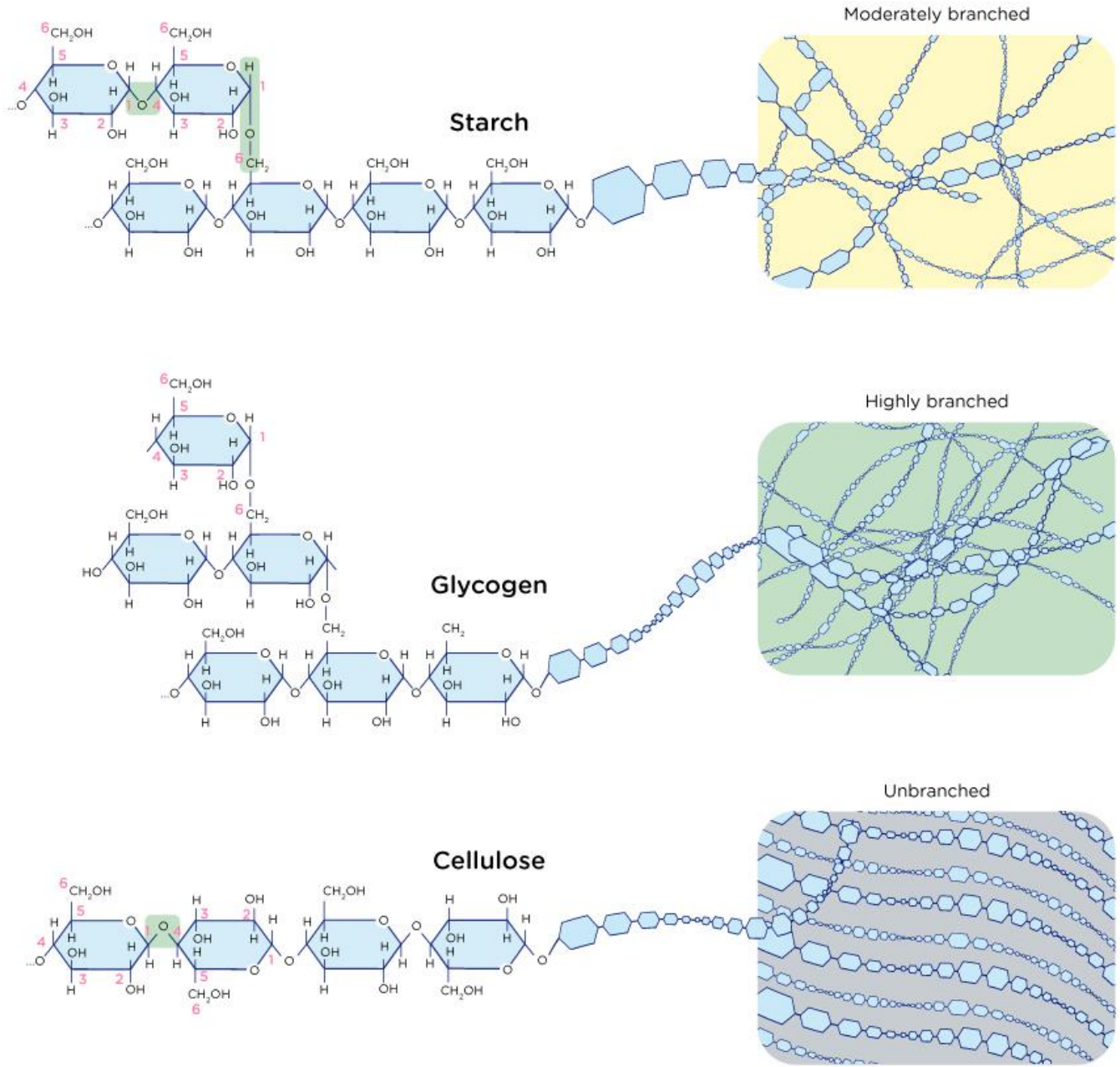




Glycogen, a polymer of glucose that is like amylopectin but more extensively branched.

Humans and other vertebrates store glycogen mainly in liver and muscle cells.

Hydrolysis of glycogen in these cells releases glucose when the demand for sugar increases.

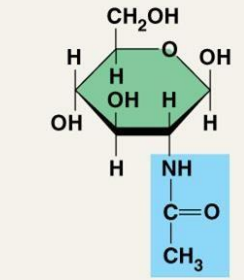

Moderately branched
Starch
Highly branched
Glycogen
Unbranched
Cellulose

Jack Westin

Chitin (Structural)

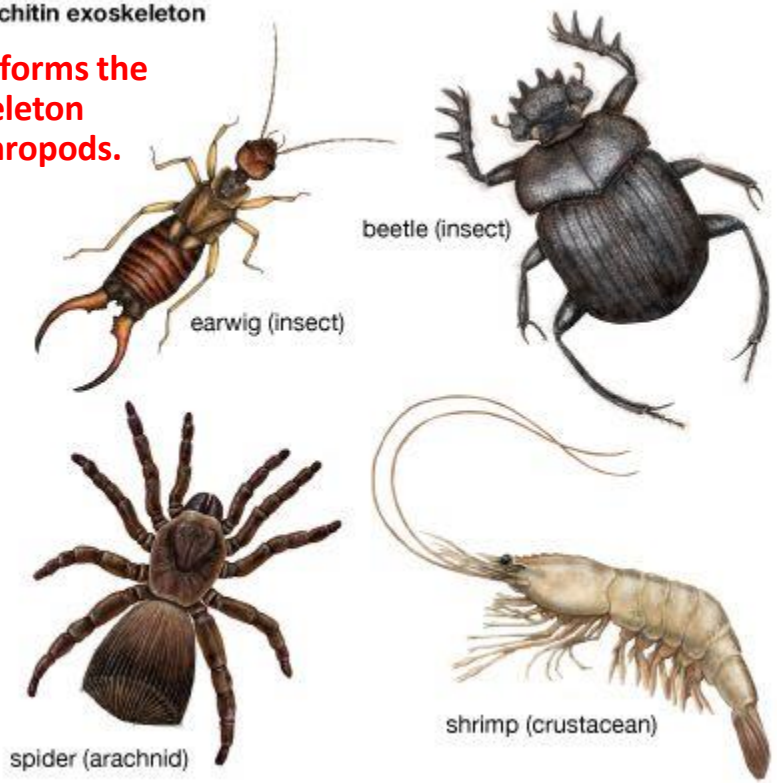
- Chitin is another structural polysaccharide, found in the exoskeleton of arthropods and many fungi.
- An exoskeleton is a hard case that surrounds the soft parts of an animal.
- Chitin is similar to cellulose, with β linkages, except that the glucose monomer of chitin has a nitrogen-containing appendage (N-acetylglucosamine).



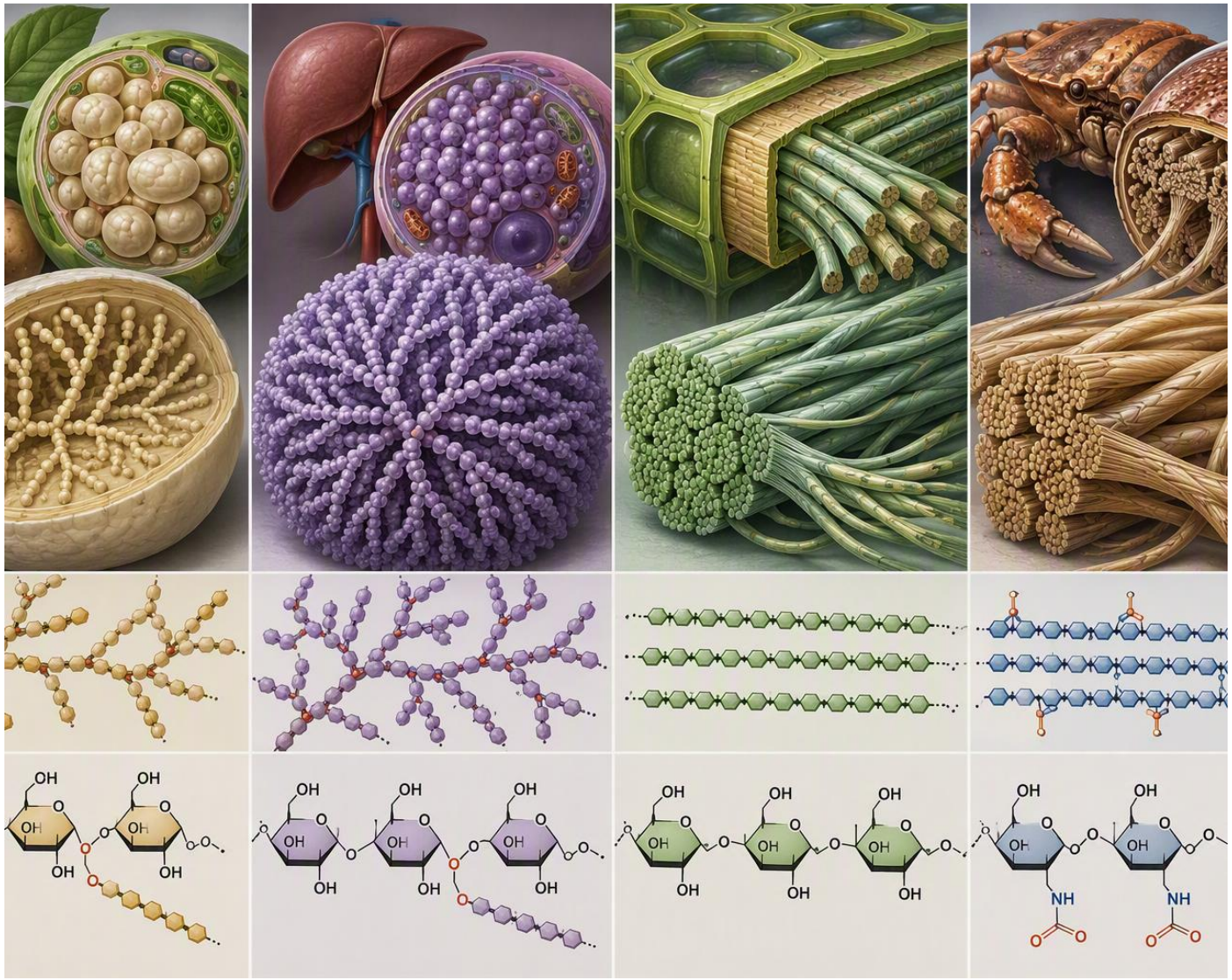


The structure of the chitin monomer

Chitin forms the exoskeleton of arthropods.



Polysaccharides: linkage controls architecture



Starch plant energy store

α(1→4), α(1→6)

Glycogen animal energy store

highly branched

Cellulose plant structural fiber

β(1→4)

Chitin structural polymer

N-acetylglucosamine

Structure → property → function

α linkages

↓

bent / helical chains

↓

enzyme-accessible

↓

energy storage

β(1→4) linkages

↓

straight chains

↓

H-bonded microfibrils

↓

tensile support

Take-home framework

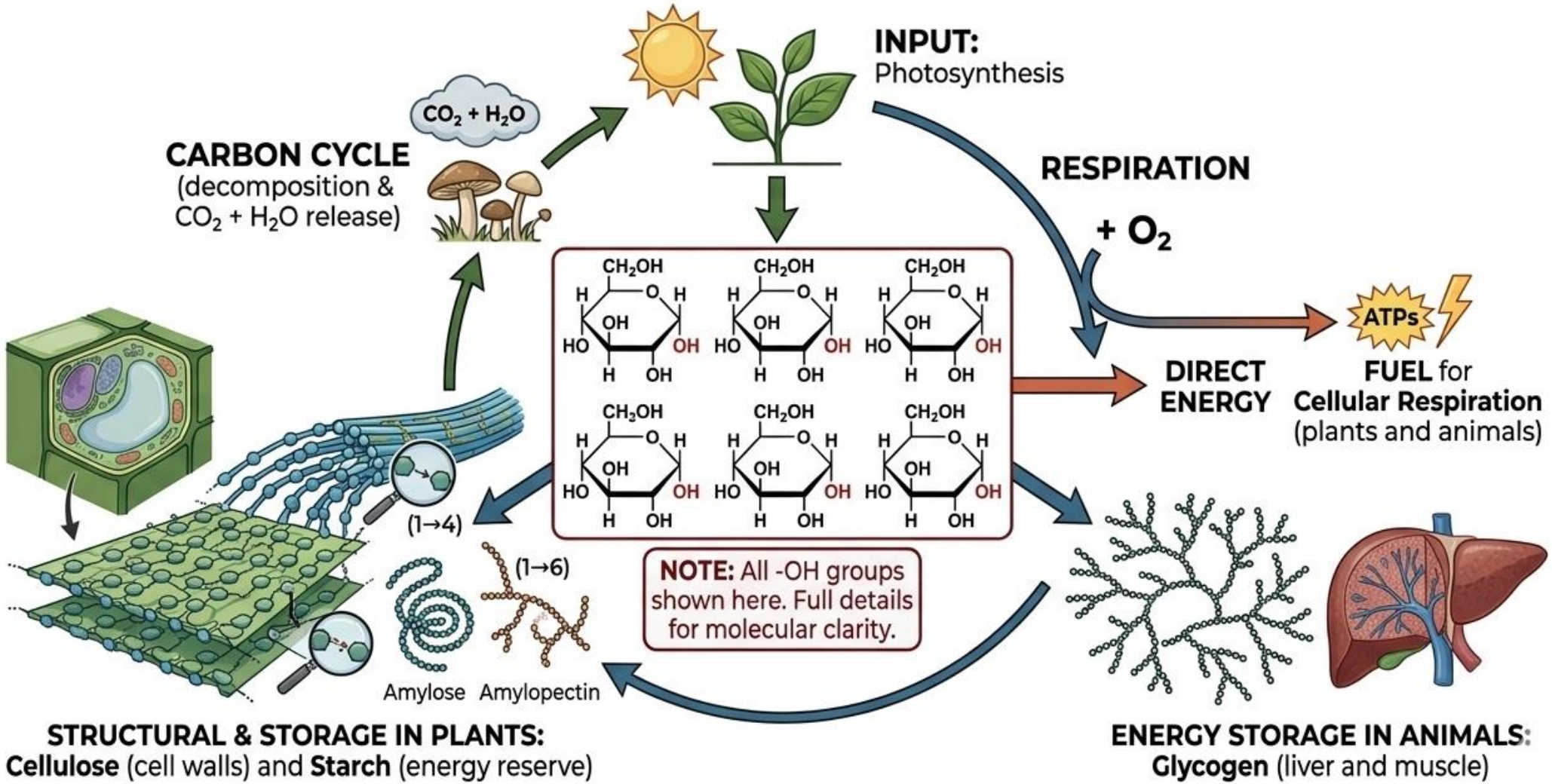
COMPOSITION	Which monosaccharides?
CONNECTIVITY	Which carbons are linked?
STEREOCHEMISTRY	α or β configuration?
ARCHITECTURE	linear, helical or branched?
FUNCTION	fuel, storage, structure or recognition?

If you can trace this chain of reasoning, you can predict carbohydrate behavior.

THE CARBOHYDRATE CYCLE: ENERGY & STORAGE.

FUNCTIONS OF CARBOHYDRATES: "GLYCANS"

Energy and energy storage – e.g. Glucose (grape sugar, dextrose)



A clear illustration of carbohydrate functions, including the full molecular structure of D-Glucose.

Too much …..
Carbohydrate will be converted into fat and stored under the skin leading to weight gain!

